

MT1008 – Multivariable Calculus

Quiz #1: Worked Solutions (BAI-2A)

Question 1 (Point-to-Plane Distance)

Find the distance between the skew lines

$$L_1 : \quad x = 1 + t, \quad y = 1 + 6t, \quad z = 2t,$$

$$L_2 : \quad x = 1 + 2s, \quad y = 5 + 15s, \quad z = -2 + 6s.$$

Solution

The direction vectors of the two lines are

$$\mathbf{v}_1 = \langle 1, 6, 2 \rangle, \quad \mathbf{v}_2 = \langle 2, 15, 6 \rangle.$$

A plane that contains L_2 and is parallel to L_1 must contain the direction vector $\mathbf{v}_2$ and be parallel to $\mathbf{v}_1$. Hence, a normal vector to this plane is

$$\mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2.$$

$$\mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 6 & 2 \\ 2 & 15 & 6 \end{vmatrix} = \langle 6, -2, 3 \rangle.$$

Choose a point on L_2 . For $s = 0$,

$$P_2 = (1, 5, -2).$$

Therefore, the equation of the plane containing L_2 and parallel to L_1 is

$$6(x - 1) - 2(y - 5) + 3(z + 2) = 0.$$

Simplifying,

$$6x - 6 - 2y + 10 + 3z + 6 = 0,$$

$$6x - 2y + 3z + 10 = 0.$$

Now choose a point on L_1 . For $t = 0$,

$$P_1 = (1, 1, 0).$$

The distance from point P_1 to the plane

$$6x - 2y + 3z + 10 = 0$$

is given by

$$d = \frac{|6(1) - 2(1) + 3(0) + 10|}{\sqrt{6^2 + (-2)^2 + 3^2}}.$$

$$d = \frac{|6 - 2 + 10|}{\sqrt{36 + 4 + 9}} = \frac{14}{7} = 2.$$

$$\boxed{d = 2}$$

Question 2

Reduce the equation to standard form and classify the surface:

$$4x^2 - y + 2z^2 = 0.$$

Solution

Rearrange the equation:

$$y = 4x^2 + 2z^2.$$

This is of the form

$$y = ax^2 + bz^2,$$

$$\boxed{\text{The given surface is an elliptic paraboloid.}}$$